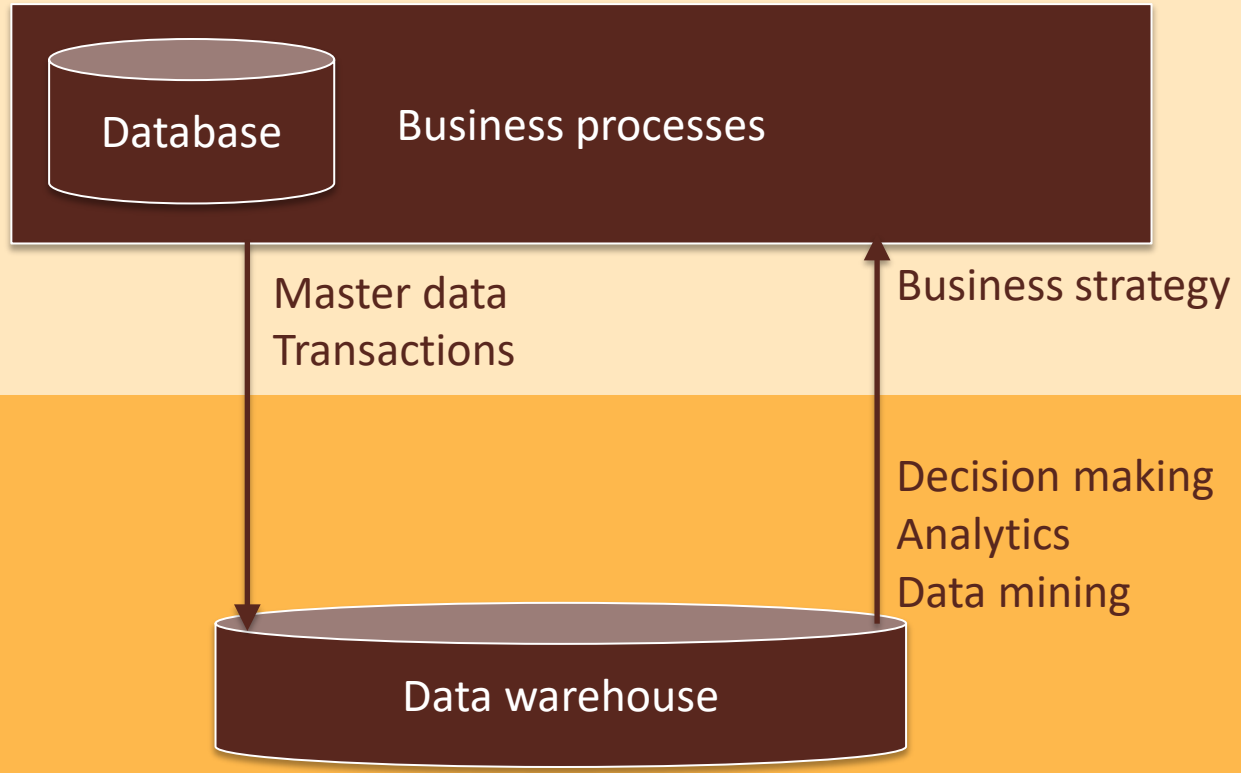
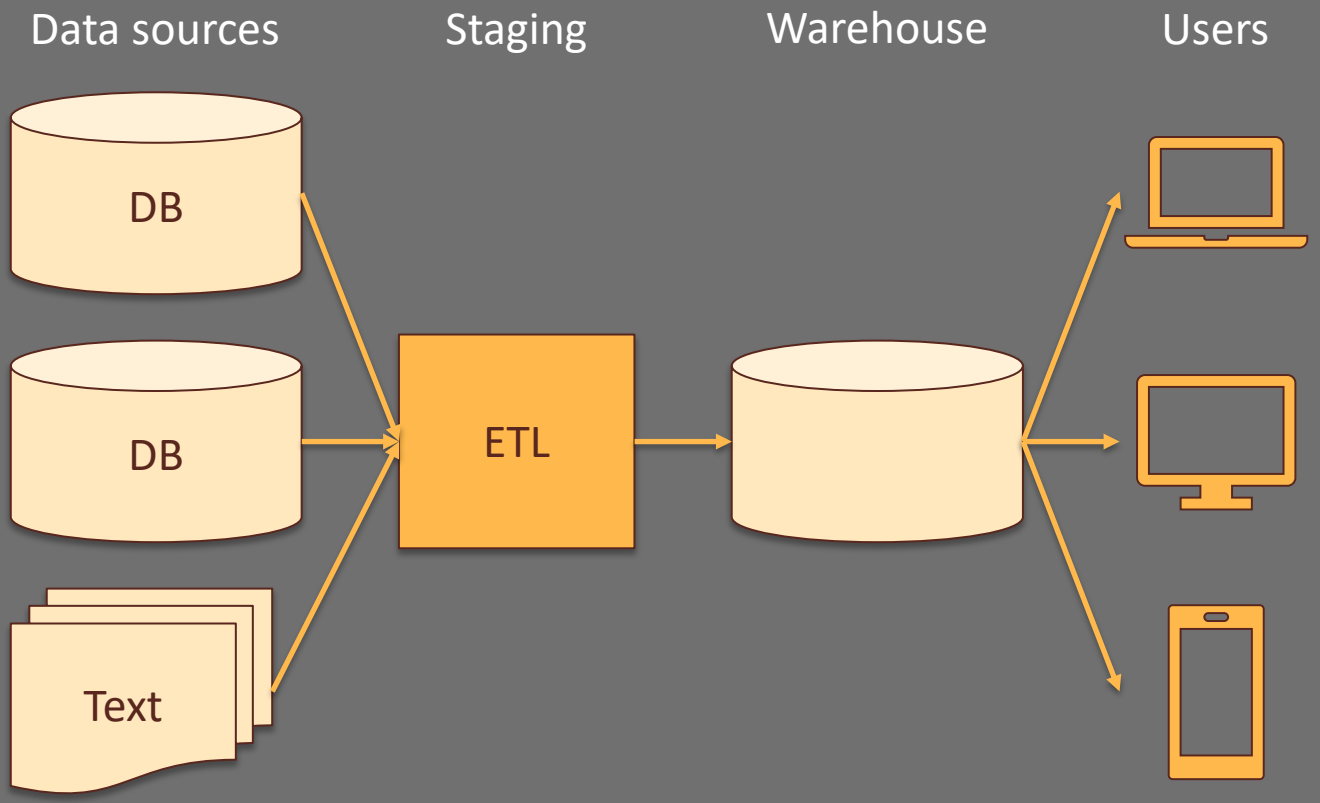
OLAP & Data Warehouses Overview

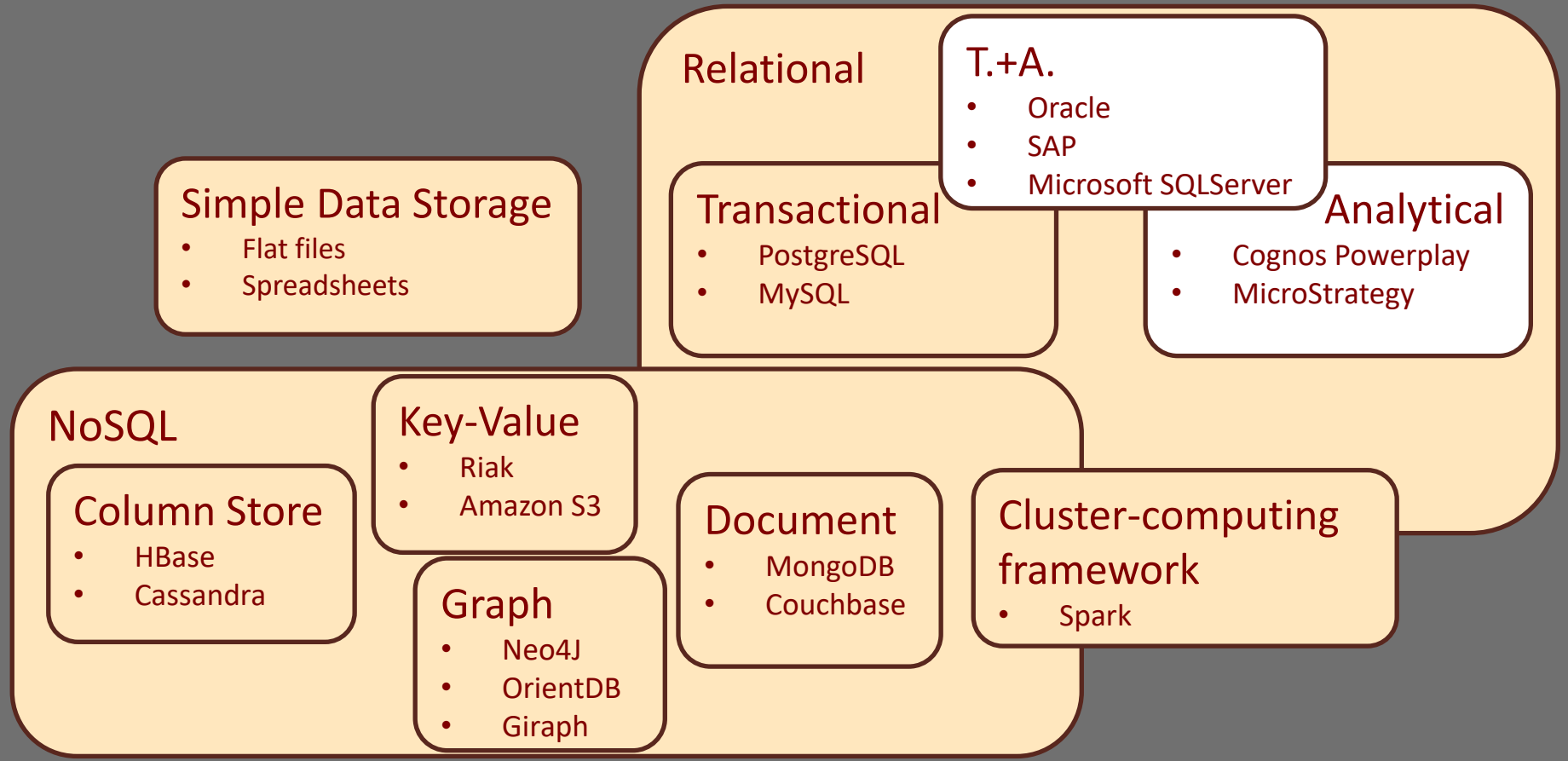
Transactional vs. Analytical

Transactional (OLTP)	Analytical (OLAP)
• E.g., retail sales	• E.g., analyze sales trends
• Track all data – CRUD	• Analyze historical data
• Frequent updates	• Periodic updates
• Flexible queries	• Predetermined queries
• Do everything well	• Optimize for specific uses




Database
Business processes
Master data
Transactions
Business strategy
OLTP - Operations
OLAP - Information
Decision making
Analytics
Data mining
Data warehouse


Data sources
Staging
Warehouse
Users
DB
DB
Text
ETL


Relational
T.+A.
Oracle
SAP
Microsoft SQLServer
Simple Data Storage
Flat files
Spreadsheets
Transactional
PostgreSQL
MySQL
Analytical
Cognos Powerplay
MicroStrategy
NoSQL
Key-Value
Riak
Amazon S3
Column Store
HBase
Cassandra
Document
MongoDB
Couchbase
Cluster-computing framework
Spark
Graph
Neo4J
OrientDB
Giraph